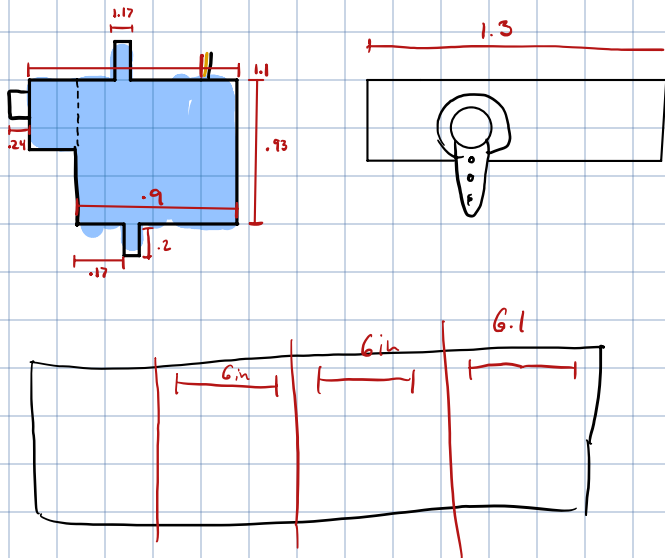


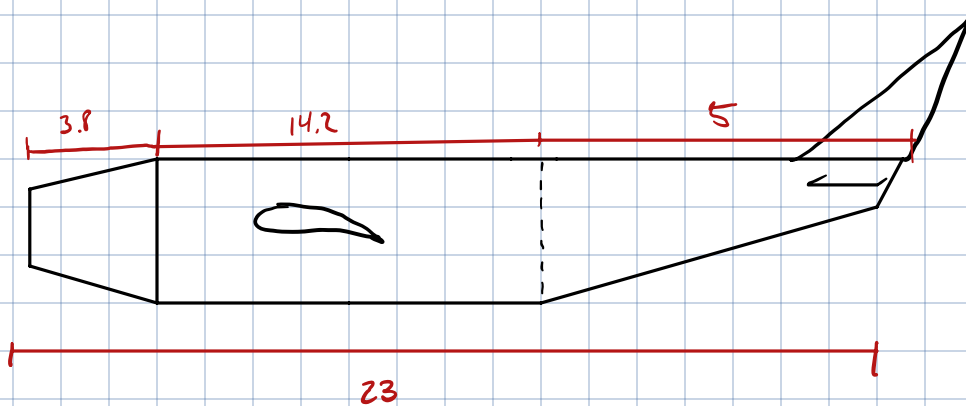
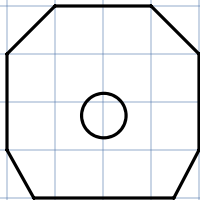
Top down view



6 $\rightarrow$ 152.4 mm

6.1 $\rightarrow$ 154.94 mm

americas hr dept @ kbr.com



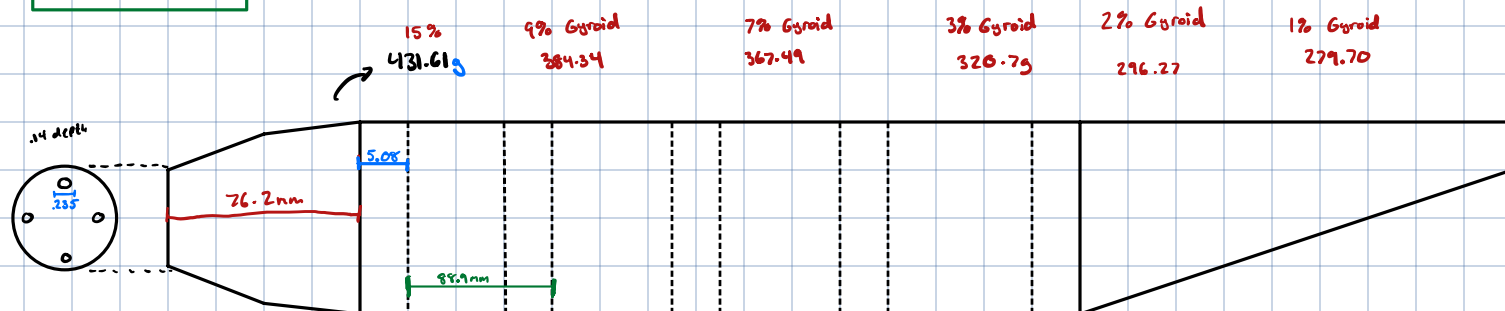
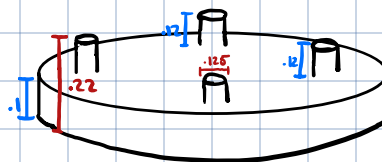
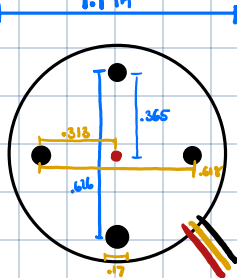
Battery
4 screws
Motor
Prop } 218g

Fuselage } 384g

Wing } 260g

862g

Weight = 8.456 N



$$L = \frac{1}{2} \rho V^2 S c_l \rightarrow \text{NACA 4412}$$

Min Takeoff Speed of 10m/s @ 4° AoA:

$$L = \frac{1}{2} (1.15 \frac{\text{kg}}{\text{m}^3}) (10 \frac{\text{m}}{\text{s}})^2 (.079 \text{ m}^2) (.846)$$

$$L = 3.842 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \rightarrow 3.842 \text{ N}$$

$$L = 3.84 \text{ N}$$

X

Min Takeoff Speed of 16m/s @ 4° AoA:

$$L = \frac{1}{2} (1.15 \frac{\text{kg}}{\text{m}^3}) (16 \frac{\text{m}}{\text{s}})^2 (.079 \text{ m}^2) (.846)$$

$$L = 9.837$$

2% infil w/ Gyroid

Takeoff with Weight = 7.595 N , AOA = 10°

$$L = \frac{1}{2} \rho V^2 S c_l$$

$$L = \frac{1}{2} (1.15 \frac{\text{kg}}{\text{m}^3}) (14 \frac{\text{m}}{\text{s}})^2 (.079 \text{ m}^2) (1.17)$$

$$L = 10.41 \text{ N}$$

Min takeoff speed = 12 m/s

↳ produces 7.65 N of Lift